Title: Clinic Appointment Scheduling and Patient No-Show Prediction System Using Machine Learning

Dataset: https://www.kaggle.com/datasets/joniarroba/noshowappointments

Student Name:

Student ID:

Table of Contents

D1 – Company Background 3

1. Organization Overview 3
2. Sector and Market 3
3. Current Challenges 4
4. Processes Needing Improvement 4
5. Strategic Opportunity 4

D2 – System Analysis 5

1. Focal Process Selection and Justification 5
2. Requirements Collection Method 5
3. Stakeholders and Roles 6
4. Detailed Tasks, Inputs/Outputs, Reports, and System Requirements 6
5. Models 7
6. Project Risk Assessment 8

D3: Design and Final Documentation 10

1. Proposed Solution Use Case Overview 10
2. DFD Level 0 11
3. DFD Level 1 11
4. UML Class Diagram 12
5. System Architecture 13

D4: Python Prototype 14

References 16

Appendix: 19

D1 – Company Background

1. Organization Overview

The Birmingham Community Healthcare NHS Foundation Trust (BCHC) is one of the foremost community-based public healthcare organizations in the United Kingdom, offering extensive community and specialist healthcare services throughout the city of Birmingham and West Midlands. The Trust provides care to thousands of patients every day with outpatient clinics, rehabilitation centers, community hospitals, as well as dental services. The mission of BCHC is to provide quality and affordable healthcare near where people live to enhance the health outcome and customer satisfaction of all age groups of patients (BMC Health Services Research, 2024). The Trust has doctors, nurses, therapists, and administrative staff teams that work together in managing the appointments of the patients and offering effective, safe, and consistent care to them.

Figure 1 Best Care Healthy Communities

Source: (Bhamcommunity.nhs.uk., 2015)

2. Sector and Market

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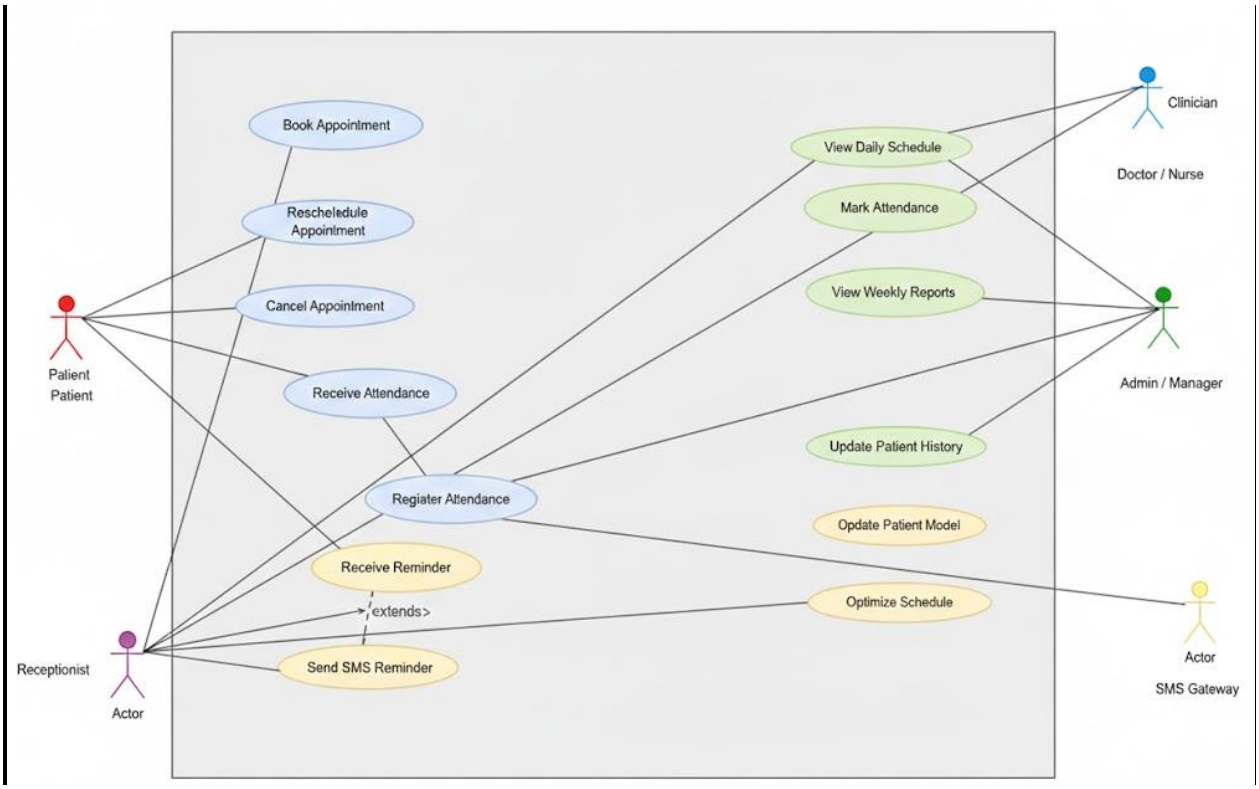

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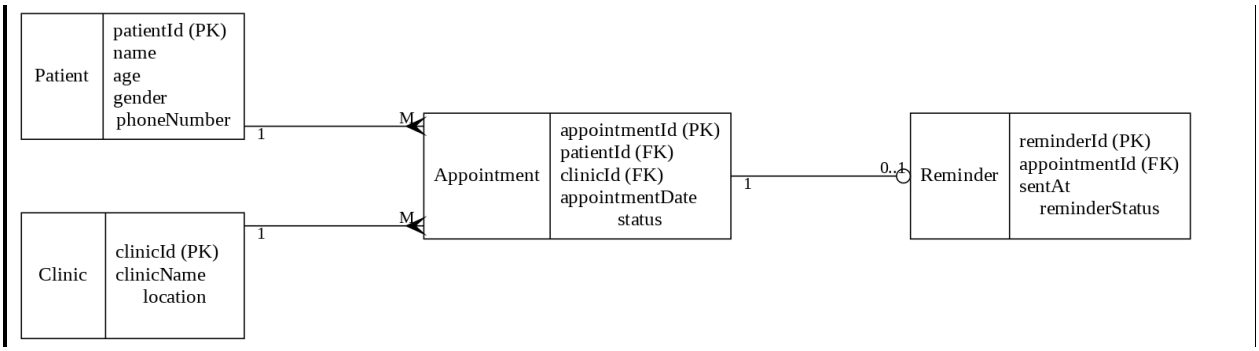

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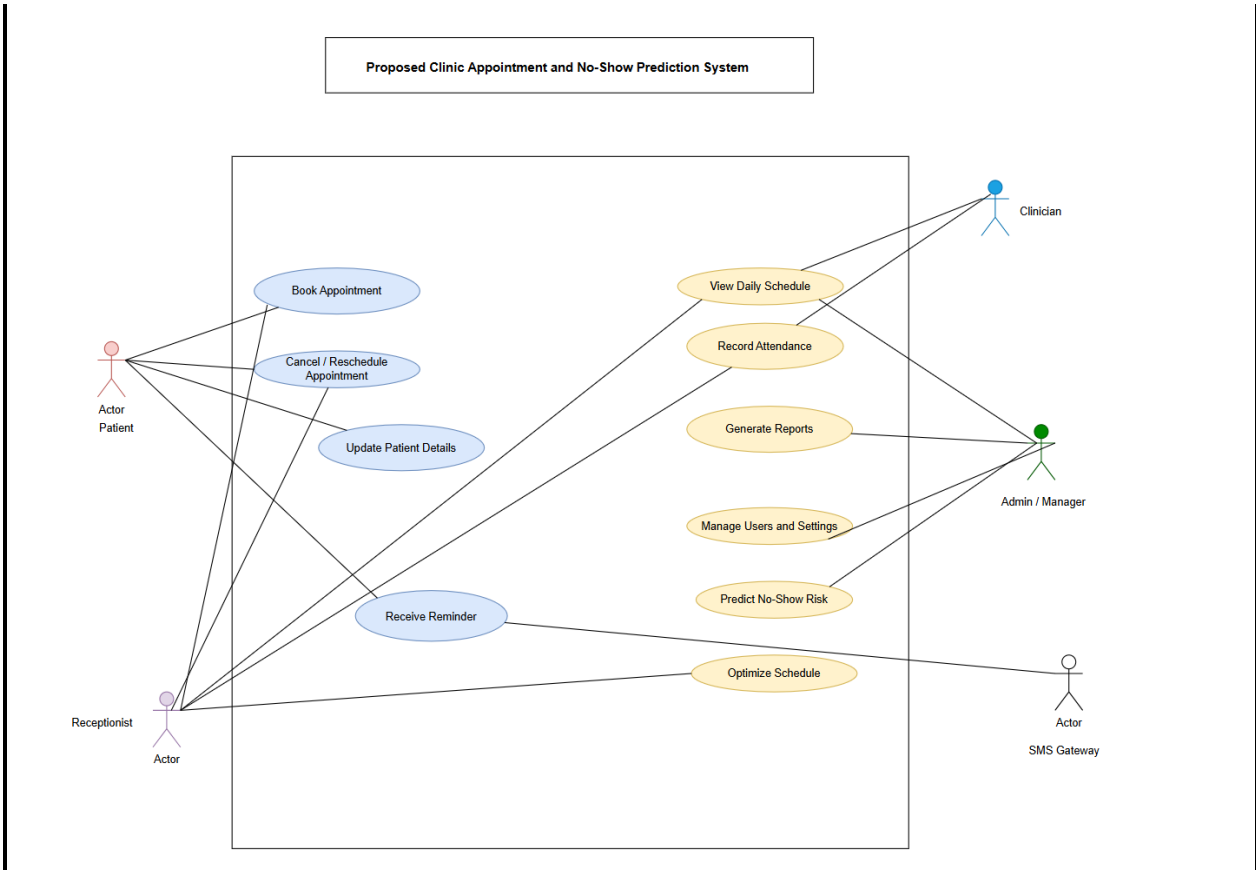

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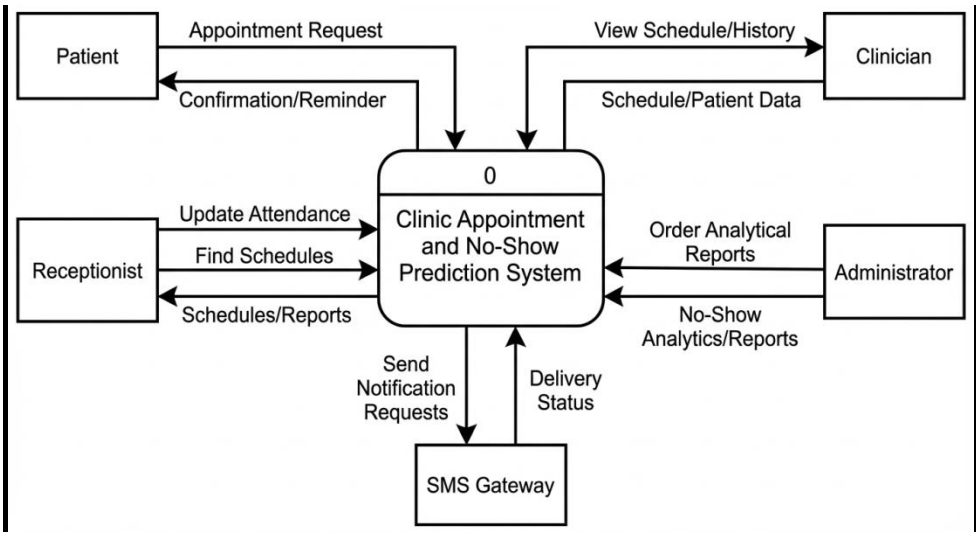

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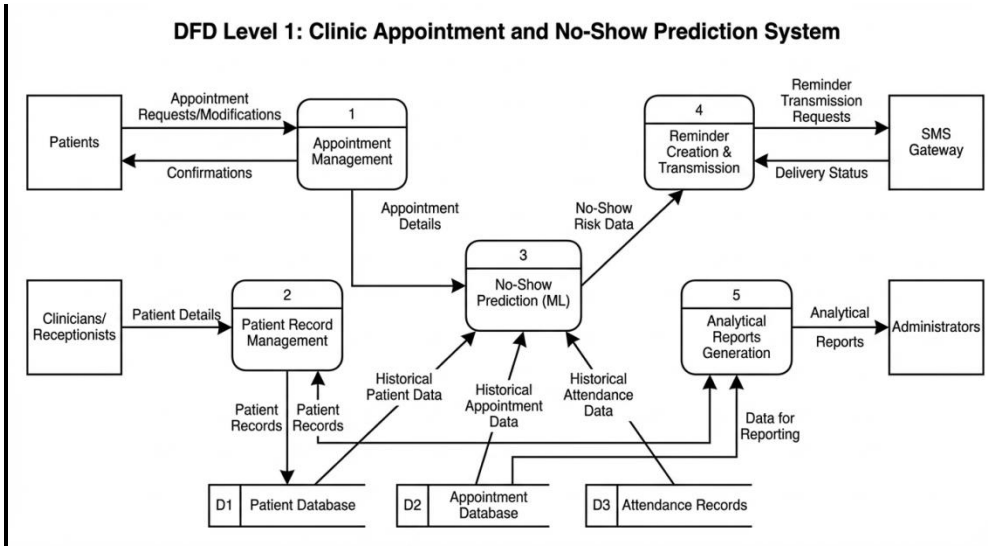

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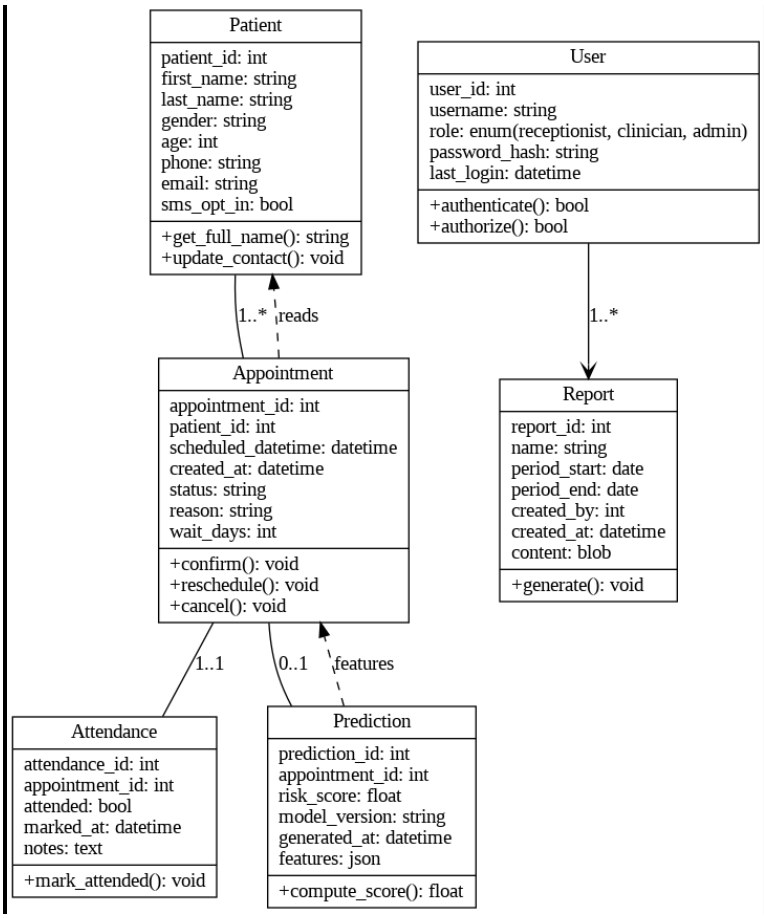

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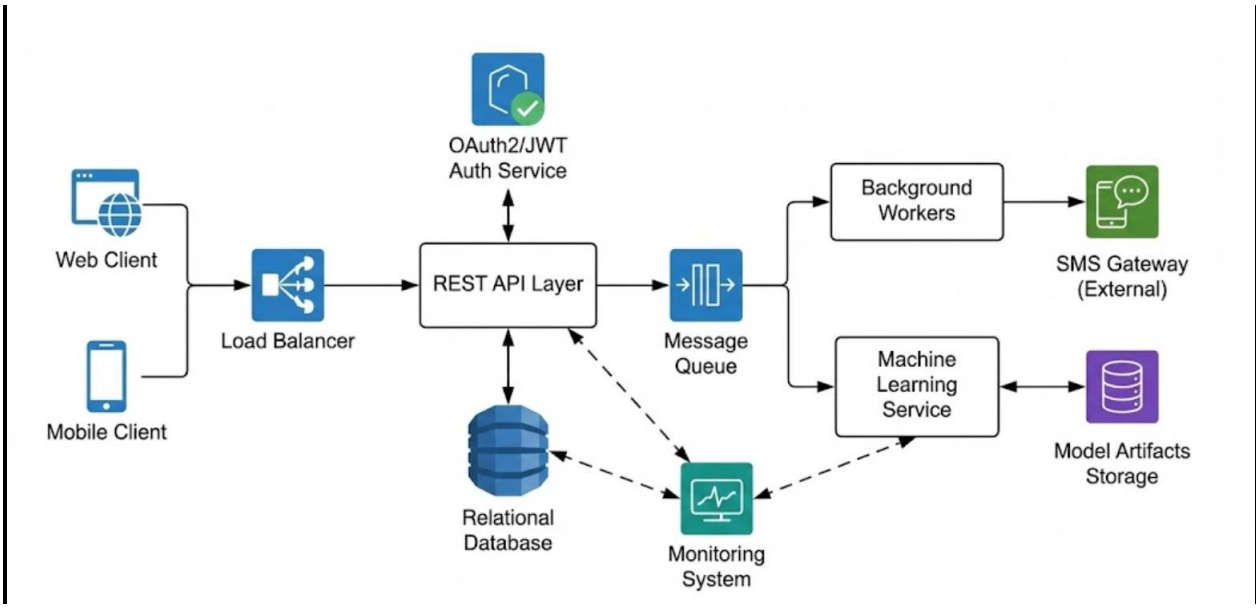

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MODEL COMPARISON TABLE

                Model  Accuracy       AUC
0  Logistic Regression  0.795853  0.660305
1        Random Forest  0.769651  0.696169
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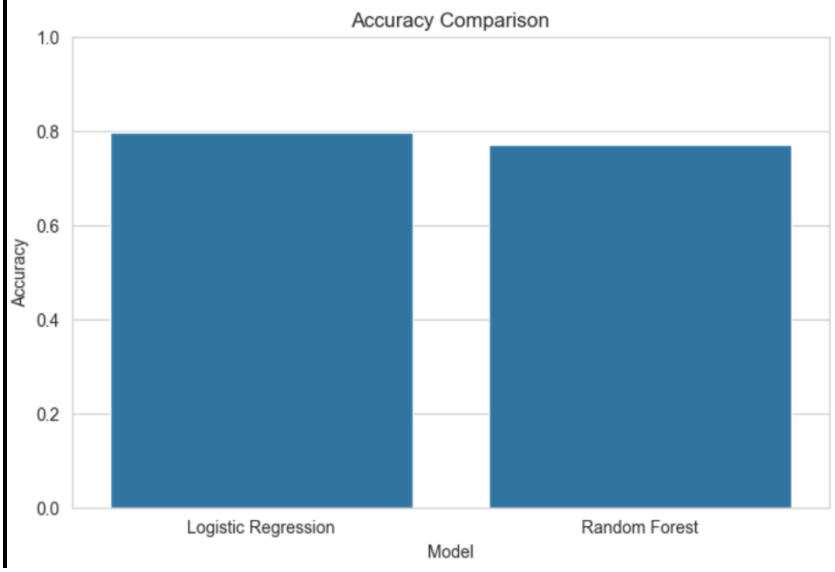

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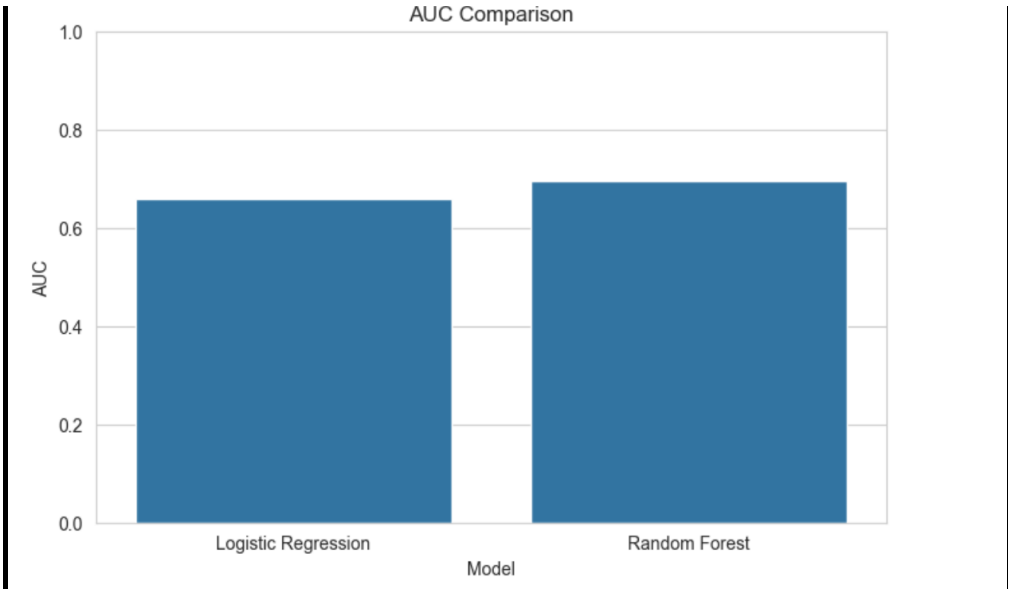

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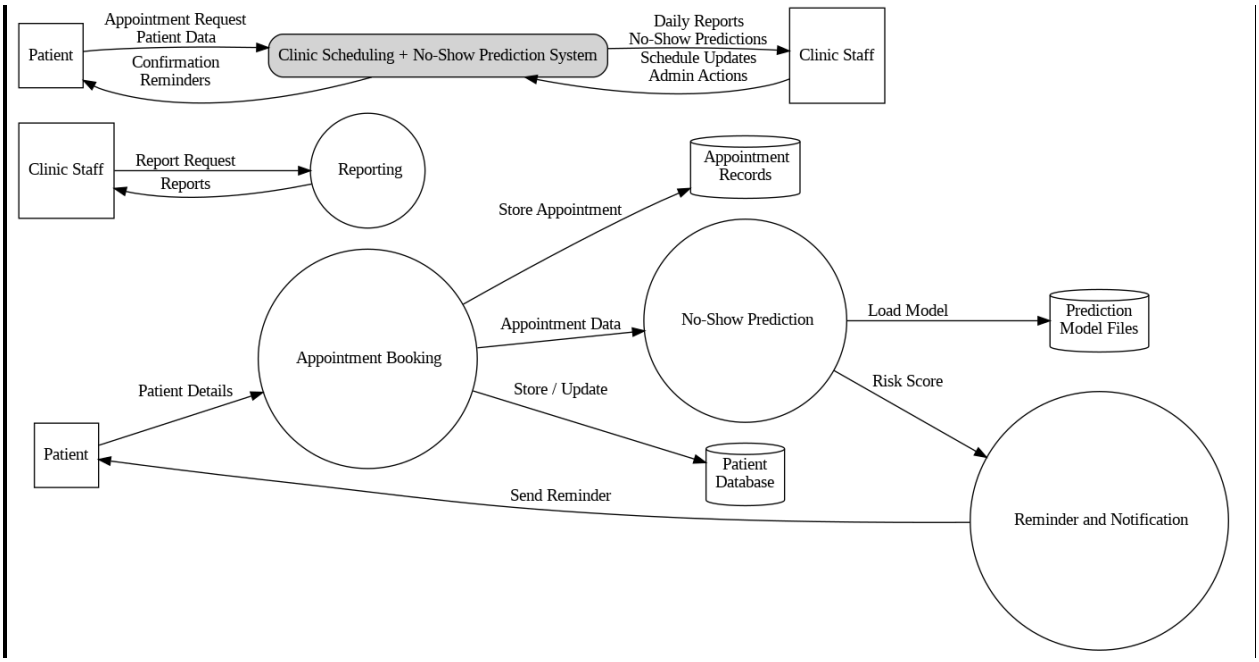

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